

Rule-Based Speech Synthesis

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In this chapter, we review some of the issues in rule-based synthesis and specifically discuss formant synthesis. Formant synthesis and the theory behind have played an important role in both the scientific progress in understanding how humans talk and also the development of the first speech technology applications. Its flexibility and small *footprint* makes the approach still of interest and a valuable complement to the current dominant methods based on concatenative data-driven synthesis. As already mentioned in the overview by *Schroeter* (Chap. 19) we also see a new trend to combine the rule-based and data-driven approaches. Formant features from a database that can be used both to optimize a rule-based formant synthesis system and to

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optimize the search for good units in a concatenative system.

20.1 Background

Rule synthesis and formant synthesis are historically highly related. The theory of formant synthesis goes back to the first attempts to model the acoustics of the speech waveform. In the publications by *Fant* [20.1, 2] and *Stevens* [20.3] the foundations of the acoustics of speech and the relation between articulatory models, acoustic manifestations, and spoken communication can be found. The synthesis models were initially evaluated by carefully copying recorded speech. After these initial experiments new methods were developed to control the synthesizer. The classic publications by *Holmes*, *Mattingly*, and *Shearme* [20.4],

Mattingly [20.5], *Flanagan* [20.6], *Klatt* [20.7], and *Allen*, *Hunnicut*, and *Klatt* [20.8] present the basic approaches for speech synthesis. An early effort to synthesize speech based on explicit rules can be found in *Carlson* and *Granström* [20.9]. Several review publications on speech synthesis have been published since then: *Sagisaka* [20.10], *Dutoit* [20.11], *Carlson* and *Granström* [20.12]. The paper by *Klatt* [20.13] gives an extensive review of the developments of the speech synthesis technique before the general introduction of concatenative synthesis.

20.2 Terminal Analog

20.2.1 Formant Synthesizers

The traditional text-to-speech systems use a terminal analog as a sound-generating device, i.e., the aim with this kind of synthesizer is only that it should be able to produce the sounds (speech spectra) that are found in natural speech. The internal structure is not a model of acoustic speech production in the vo-

cal tract. The basic concept is the combination of sound sources and filters, describing the transfer function, see *Lawrence* [20.14] and *Fant* [20.15]. The source-filter model exists in several versions; the classical configuration by *Klatt* [20.16], is exemplified in Fig. 20.1. The vocal-tract transfer function is simulated by a sequence of second-order filters in a cascaded arrangement, while a parallel structure is used mostly

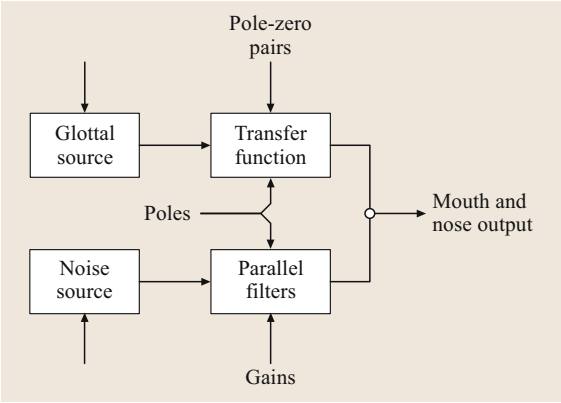


Fig. 20.1 Block diagram of the main components of the Klatt terminal-analog speech synthesizer (after [20.18])

for the synthesis of consonants. One important advantage of a cascade synthesizer is the automatic setting of formant amplitudes. The disadvantage is that it sometimes can be difficult to do detailed spectral matching between natural and synthesized spectra because of the simplified model. Parallel synthesizers such as the one by *Holmes* [20.17] do not have this limitation.

The Klatt model is widely used in research both for general synthesis purposes and for perceptual experiments. A simplified version of this system is used in commercial products that builds on the research at Massachusetts Institute of Technology (MIT): MITalk [20.8], Klattalk [20.20], and DECTalk [20.21].

The formant terminal analog, GLOVE [20.19], based on the OVE synthesizer [20.22], was developed at KTH (Royal Institute of Technology) and has been

explored in speech research and several practical applications [20.23, 24]. The structure of the GLOVE synthesizer is shown in Fig. 20.2. The controllable parameters are indicated by two-letter symbols. To the left, the two sound sources can be seen. For mixed excitation the sources are connected in two ways. The parameter N_M flow-modulates the noise source, typical for voiced fricatives. The parameter N_A adds noise to the glottal source, as in breathy or whispered voices. The five parameters above the voice source are the glottal parameters referred to in the voice source section below. The sound source signals are fed into the three parallel branches with poles and zeroes. All are controlled by amplitude parameters (A_N , A_0 , A_H and A_C). The upper branch is primarily used for introducing an extra pole and zero in nasals and nasalized sounds. The middle branch is the main branch for sounds produced with glottal excitation and the lowest branch models sounds with supraglottal excitation, such as stops and fricatives. This basic configuration can be augmented in several ways. The interaction between the source and the vocal tract, which can be substantial, is in this case only modeled by the B_M parameter that modulates the bandwidth of the first formant, dependent on the glottal opening, or more precisely the glottal flow. The main difference between the MIT and KTH traditions can be found in how the consonants are modeled. In the OVE case, a fricative is filtered by a zero–pole–pole configuration rather than the parallel branch in the Klatt synthesizer.

With terminal analog synthesizers, it is possible to simulate most human voices, and to replicate an utterance without noticeable quality reduction. However, it is interesting to note that some voices are easier to model than others. Despite the progress, speech quality is not good enough in all applications of text to speech. The main reasons for the limited success in formant-based synthesis can be explained by the incomplete knowledge needed for automatic control of the parameters. It should be noted that the transfer of knowledge from phonetics to speech technology and explicit rule-based description has not been an easy process.

20.2.2 Higher-Level Parameters

Since the control of a formant synthesizer can be a very complex task, some efforts have been made to help the developer. The *higher-level parameters* described by *Stevens* and *Bickley* [20.18], *Hanson* and *Stevens* [20.25] and *Stevens* [20.26], for example, explore an intermediate level that is more understand-

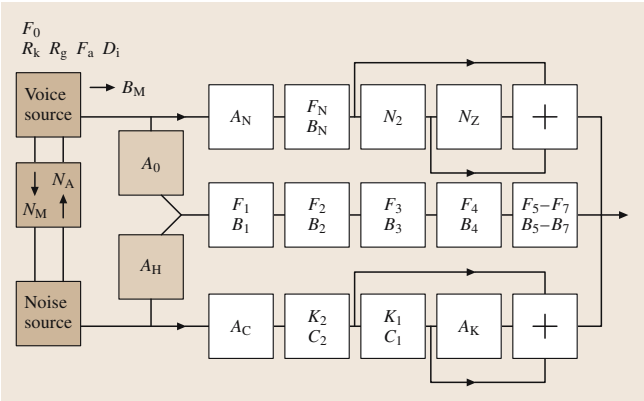


Fig. 20.2 Block diagram of the main components of the terminal-analog speech synthesizer GLOVE (after [20.19])

able from the developer's point of view compared to the detailed synthesizer specifications. The goal with this approach is to find a synthesis framework to simplify the process and to incorporate the constraints that are known to exist within the process. A formant frequency should not have to be adjusted specifically by the rule developer depending on nasality or glottal opening. This type of adjustment might be better handled automatically according to a well-specified model. The same process should occur with other parameters such as bandwidths and glottal settings. This approach requires detailed understanding of the relation between acoustic and articulatory phonetics. The Prosynth project Ogden et al. [20.27] has made use of the multilevel view and included phonetic and prosodic aspects in the approach; see Heid and Hawkins [20.28].

20.2.3 Voice Source Models

The traditional voice source model has been a simple or double impulse. This is one reason why the voices produced by early text-to-speech systems lack naturalness to a great extent. While the male voice has sometimes been regarded as generally acceptable, an improved glottal source will open the way to more-realistic synthesis of child and female voices and also to greater naturalness and variation in male voices.

Most source models work in the time domain with various controls to manipulate the pulse shape [20.30–40]. One influential voice source model is the LF model [20.41]. It has a truncated exponential sinusoid followed by a variable-cut-off -6 dB/octave low-pass filter modeling the effect of the return phase, i. e., the time from maximum excitation of the vocal tract to complete closure of the vocal folds. Figure 20.3 shows the function of the control parameters. In addition to the amplitude and fundamental frequency control, two parameters largely influence the amplitudes of the two to three lowest harmonics, and one parameter, the high-frequency content of the spectrum. Another vocal source parameter is the diplophonia parameter (inspired by Klatt and Klatt [20.33]) with which creak, laryngalization or diplophonia can be modelled. This parameter influences the function of the voiced source in such a way that every second pulse is lowered in amplitude and shifted in time.

The acoustic interactions between the glottal source and the vocal tract also have to be considered [20.42]. One of the major factors in this respect is the varying bandwidth of the formants. This is especially true for

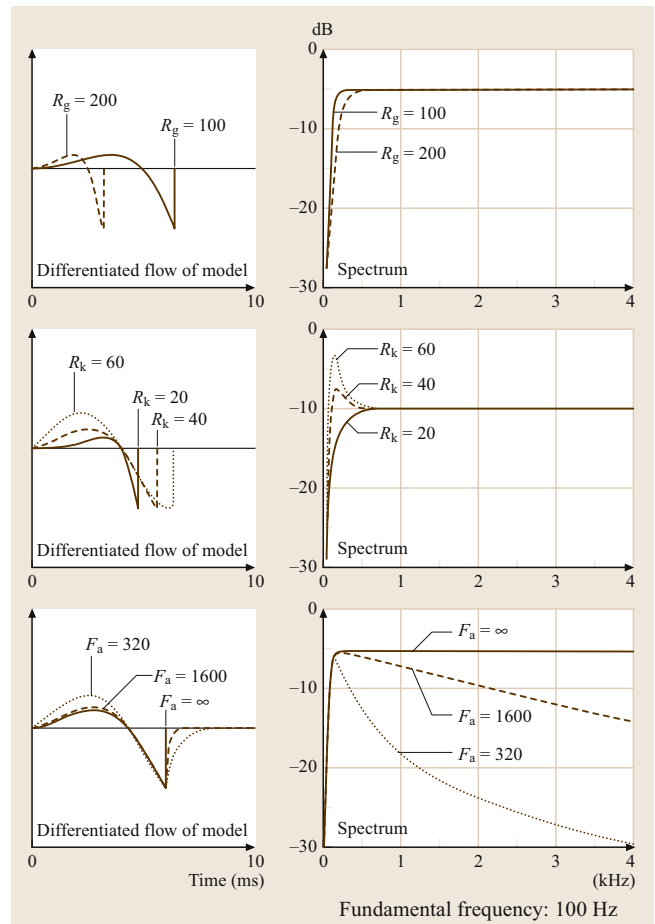


Fig. 20.3 The influence of the parameters R_G , R_K and F_A on the differentiated glottal flow pulse shape and spectrum. The spectra are pre-emphasized by 6 dB/oct (after [20.29])

the first formant, which can be heavily damped during the open phase of the glottal source. However it is not clear that such a variation can be perceived by a listener. Listeners tend to be rather insensitive to bandwidth variation [20.6]. When more-complex models are included, the output from the model has to change from a glottal flow model to a model of the glottal opening. The subglottal cavities can then be included in an articulatory model.

Noise sources have attracted much less research effort than the voiced source. However, some aspects are discussed by Stevens [20.3, 43], Shadle [20.44], and Badin and Fant [20.45]. Typically, simple white noise is filtered by resonances that are stationary between each parameter frame. Some formant synthesizers do include

some interaction between the voice source and the noise source, but the interaction is rather primitive. The mod-

eling of transient sounds and aspiration dependent on vocal-cord opening is still a challenge.

20.3 Controlling the Synthesizer

As already mentioned initial work on formant synthesis was related to the development of theories and models of the acoustics of speech. Synthesis experiments were mostly devoted to testing the models and initial speech perception experiments, for example the classic experiments by *Delattre, Liberman, and Cooper* [20.47]. The work by *Holmes, Mattingly and Shearme* [20.4] was a major step towards controlling the synthesizer based on symbolic input such as phones or phonemes. Each phoneme was described by a simplified parametric pattern. It was clear that quality was limited if the settings were not adjusted, and rule systems to modify these settings were developed. A number of models were created to take coarticulation into account such as smoothed step functions *Liljencrants* [20.48] or explicit rule descriptions *Klatt* [20.9, 49–51].

Unfortunately rule development for formant synthesis has some problems due to the parametric domain itself. The context-dependent acoustic realizations in terms of formant settings reflect the underlying articulatory gestures rather than the formant pattern in the context (e.g., [20.52]). In a similar way parameters such as variation along the *hyper/hypo-articulation* dimension have a strong influence on the acoustic realization and talking speed [20.53, 54].

A challenging problem in rule development is the locus equations. For example velar stop coarticulation is dependent on many factors and cannot easily be summarized in a single formula [20.55].

These are some of the reasons why formant synthesis has not been competitive in relation to concatenative synthesis. However, the need to synthesize different voices and voice characteristics and to model emotive speech is a strong motivation to keep research on parametric synthesis active. The driving force is that rule-based formant synthesis has the necessary flexibility to model both linguistic and extralinguistic processes. It is, for example, evident that large databases for corpus-based approaches cannot be recorded consistently in a range of desired speaking styles. We will in the following describe some tools that have been used to develop formant synthesis rules and alternative data-driven methods. Rules for prosodic realization will be

discussed further in the contribution by Van Santen et al. (Chap. 20).

20.3.1 Rule Compilers for Speech Synthesis

Development tools for text-to-speech systems have received considerable attention. Often the basis of such tools has followed the development of phonological theory. The work on generative phonology and especially the publication of *The Sound Pattern of English* by *Chomsky and Halle* [20.56] led to a special kind of synthesis system based on context-sensitive rewrite rules. The linguistically oriented notation inspired speech researchers to create special rule compilers for text-to-speech developments in the 1970s (e.g. [20.36]). The implementations vary depending on the developer’s inclination. It is important to note that crucial decisions were often hidden in the systems. The rules may operate rule-by-rule or segment-by-segment. With greater emphasis on prosodic modeling and the related development of nonlinear phonology [20.57], synthesis procedures inspired by such theories were created, as in the systems described by *Hertz* [20.58], *Hertz, Kadin, and Karplus* [20.59], *Lazzaretto and Nebbia* [20.60], *Ceder and Lyberg* [20.61], and *Leeuwen and Lindert* [20.46, 62]. The common feature of these notations was that they keep information on different linguis-

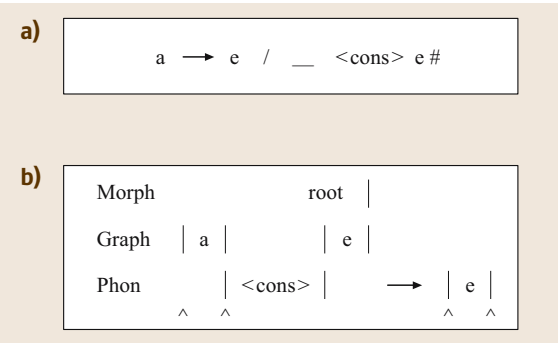


Fig. 20.4 (a) Notation according to the KTH RULSYS system. (after [20.9]) (b) Notation according to the *Speech Maker* system developed at IPO (after [20.46])

tic levels (tiers) separate in a more-explicit way than the essentially linear representation based on generative phonology. This gave potentially higher flexibility, but also more-complex notations. The rules now looked more like two-dimensional schemes rather than one-line representations. This posed greater demands on the developer and his/her tools. Rather than conventional text editors, editors should preferably be graphic editors, where relations between tiers can easily be specified, as in the Speech Maker system developed at IPO (Institute for Preception Research) [20.46]. In Fig. 20.4, this representation is compared to the one used in the KTH system. The rules describe how a letter “a” is pronounced as [e] if it occurs before a single consonant sound and a root final “e”. In the KTH notation, the root boundary is introduced as the symbol “#” by earlier rules. After the rule application, the letter origin of the [e] is lost, which places greater demands on the rule order.

20.3.2 Data-Driven Parametric Synthesis

The flexibility in formant synthesis can also be a problem, since articulatory constraints, for example, are not directly included in the formant-based model. The underlying articulatory gestures are not easily transformed to the acoustic domain described by the formant model. When increasing our ambitions to multilingual, multispeaker and multistyle synthesis it is obvious that we want to find at least semiautomatic methods to collect the necessary information, using speech and language databases. Traditionally, speech synthesis has been based on very labor-intensive optimization work. The notion *analysis by synthesis* has not been explored except by manual comparisons between hand-tuned spectral slices and a reference spectrum. The work by Holmes and Pearce [20.64] is a good example of how to speed up this process. With the help of a synthesis model, the spectra are automatically matched against analyzed speech. Automatic techniques such as this will also probably play an important role in making speaker-dependent adjustments. One advantage of these methods is that the optimization is done in the same framework as that to be used in the production. The synthesizer constraints are thus already imposed in the initial state.

As early as in the 1950s Peterson et al. [20.65] suggested that unit concatenation might be a possible solution for speech synthesis. Dixon and Maxey [20.66]

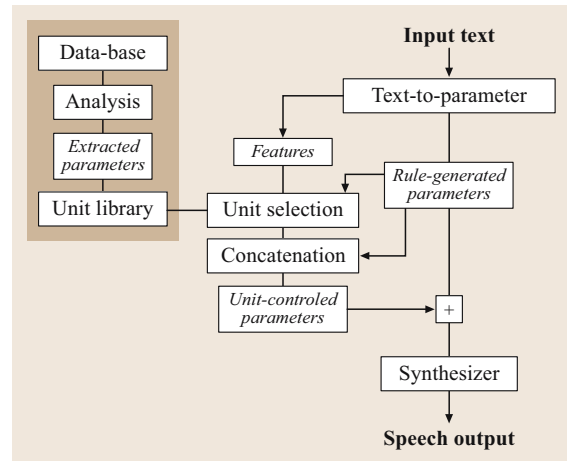


Fig. 20.5 Rule-based synthesis system using a data-driven unit library (after [20.63])

made a special effort to create a unit library for diphone synthesis. Early synthesis research at AT&T based on *diadic units* [20.67] demonstrated an alternative to rule-based formant synthesis.

These alternative approaches, which reduce the need for detailed formant synthesis rules but still keep the flexibility of the formant model, extract formant synthesis parameters directly from a labeled corpus. Mannell [20.68] reported a promising effort to create a diphone library for formant synthesis. The procedure included a speaker-specific extraction of formant frequencies from a labeled database. In a sequence of papers from Utsunomiya University, Japan automatic formant tracking was used to generate speech synthesis of high quality using formant synthesis and an elaborate voice source (e.g., Mori et al. [20.69]). Research efforts to combine data-driven and rule-based methods has recently been reported by Hertz [20.70], and Carlson and Granström [20.63]. The approaches take advantage of the fact that a unit library can better model detailed gestures than general rules. Figure 20.5 illustrates the approach in the KTH text-to-speech system. A database is used to create a unit library and the library information is mixed with the rule-driven parameters. Automatic methods of formant extraction are of course preferred when creating a unit library, due to the amount of data that has to be processed [20.71–73]. However, available methods do not always perform adequately and manual editing is often necessary.

20.4 Special Applications of Rule-Based Parametric Synthesis

In some cases where a speaker model is not readily available database-driven concatenative synthesis is not so well suited. One such example that we have been involved in is the experimental synthesis of varieties of deaf speech in the context of speech training [20.74, 75]. The aim in this project was to synthesize some prototypical versions of the deviations found in the speech of several deaf children, like consonantal realizations, reduced and distorted vowel inventory, and deviant prosody. The rule-based method made it easy to synthesize and evaluate different versions with fewer deviations, thus simulating the effect of (successful) speech training. The often-systematic differences that make intelligibility difficult for occasional listeners, tend to affect the understanding of persons like family less. This points to another use of speech synthesis, where the deviant speech could be *corrected*, but still retain the voice of the deaf or speech-impaired person. One recent example of work aimed at dysarthric speakers

is described in Kain et al. [20.76], where some improvement was demonstrated both in intelligibility and quality by formant analysis, modification, and resynthesis. The experiment concerned only vowels in nonsense CVC (consonant-vowel-consonant) words uttered by one dysarthric speaker, but points to an interesting possibility.

One early application of speech synthesis was as speech prosthesis for nonvocal individuals. Ideally such a device should be able to use a variety of speaking styles, displaying different attitudes, and emotions in a voice selected by the user. This makes conventional unit-selection methods less suitable, due to the difficulty of recording an appropriate database. Several attempts have made to construct such a prosthesis, e.g., Murray et al. [20.77], and Cudd et al. [20.78], or the components necessary for, e.g., emotion control [20.79]. However, presently commercially available speech prostheses still lack most of the desired functions.

20.5 Concluding Remarks

Recently we have also seen renewed commercial interest in speech synthesis using the formant model [e.g., the Aurix text-to-speech (TTS) system from 20/20 Speech]. One motivation is the need to generate speech using a very small *footprint*. Thus, one

can predict that formant synthesis will again be an important research subject because of its flexibility and also because of how the formant synthesis approach can be compressed into a limited application environment.

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